ASSIGNMENT 3
Apply Dimensionality Reduction Using PCA

1. Aim

To reduce the number of features in a dataset using Principal Component Analysis (PCA).

2. Objectives

1. To understand dimensionality reduction.
2. To learn the purpose of principal components.
3. To apply PCA to numerical data.
4. To reduce the number of features while retaining as much variation as possible.
5. To visualise high-dimensional data in two dimensions.

3. Theory

A dataset may contain many features or attributes. When the number of features becomes large, data processing can require more time and memory. It can also become difficult to visualise the data.

Dimensionality reduction is the process of reducing the number of dimensions or features used to represent a dataset while attempting to preserve important information.

Principal Component Analysis (PCA) is a commonly used dimensionality-reduction technique. It transforms the original numerical features into new features called principal components.

The first principal component captures the largest possible amount of variation in the data. The second principal component captures as much of the remaining variation as possible while remaining perpendicular to the first component.

PCA does not simply select two original columns. Instead, it creates new combinations of the original features.

In this experiment, the Iris dataset contains 150 flower observations and four numerical features. PCA reduces the four features to two principal components, allowing the observations to be displayed on a two-dimensional graph.

Before applying PCA, the features are standardised because features with larger numerical scales can otherwise have a greater influence on the result.

The explained variance ratio indicates how much of the dataset's variation is represented by each principal component.

4. Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Display the original dataset dimensions.
4. Standardise the features.
5. Apply PCA with two principal components.
6. Display the reduced dimensions.

5. Explanation of Important Code

- load_iris() loads a dataset containing 150 flower observations and four features.
- StandardScaler() standardises the features.
- PCA(n_components=2) specifies that the reduced dataset should contain two principal components
- fit_transform() learns the principal components and transforms the original data.
- explained_variance_ratio_ shows the proportion of variation represented by each component.
- plt.scatter() displays the reduced data on a two-dimensional graph.

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6. Applications

1. Visualising datasets containing many features.
2. Reducing storage and computational requirements.
3. Simplifying numerical datasets before further analysis.
4. Analysing image and sensor data.
5. Exploring patterns in biological and scientific datasets.

7. Advantages

1. Reduces the number of dimensions.
2. Exploring patterns in biological and scientific datasets.
3. Can reduce computational requirements for later processing.
4. Combines correlated numerical features into fewer components.

8. Limitations

1. Principal components may be difficult to interpret.
2. Some information may be lost when dimensions are removed.
3. PCA primarily captures linear patterns.
4. Standardisation and appropriate preprocessing are important.

9. Conclusion

PCA was applied to reduce the Iris dataset from four numerical features to two principal components. The reduced dimensions and explained variance ratios demonstrated how dimensionality reduction can simplify data while preserving a substantial portion of its variation.